



Federal Urdu University
of Arts, Science and Technology

UNIVERSITY EVENT PLANNING AGENT (UNIVENTO)

Course Project: Artificial Intelligence

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UNIVERSITY EVENT PLANNING AI AGENT

1. Identify your problem in the context of agent.

University events such as seminars, workshops, competitions, and cultural programs require proper planning of venue, time, budget, participants, and resources. Organizers have to manage these things manually, which is time-consuming and can lead to scheduling conflicts, missing resources, or delays.

To solve this problem, a University Event Planning AI Agent has been designed that works on predefined condition-action rules. This agent checks the current event requirements and reacts according to a set of fixed rules.

For example: It can suggest a suitable venue, check the available resources, remind organizers about upcoming tasks, and identify scheduling conflicts.

2. Problem Formulation

Textual Representation

Sensor:

The agent collects information about the event, including event type, date, time, number of attendees, venue availability, budget, required resources, and pending tasks.

Actuator:

The agent performs actions such as recommending an appropriate venue and time, sending reminders, showing event details, and providing useful planning suggestions.

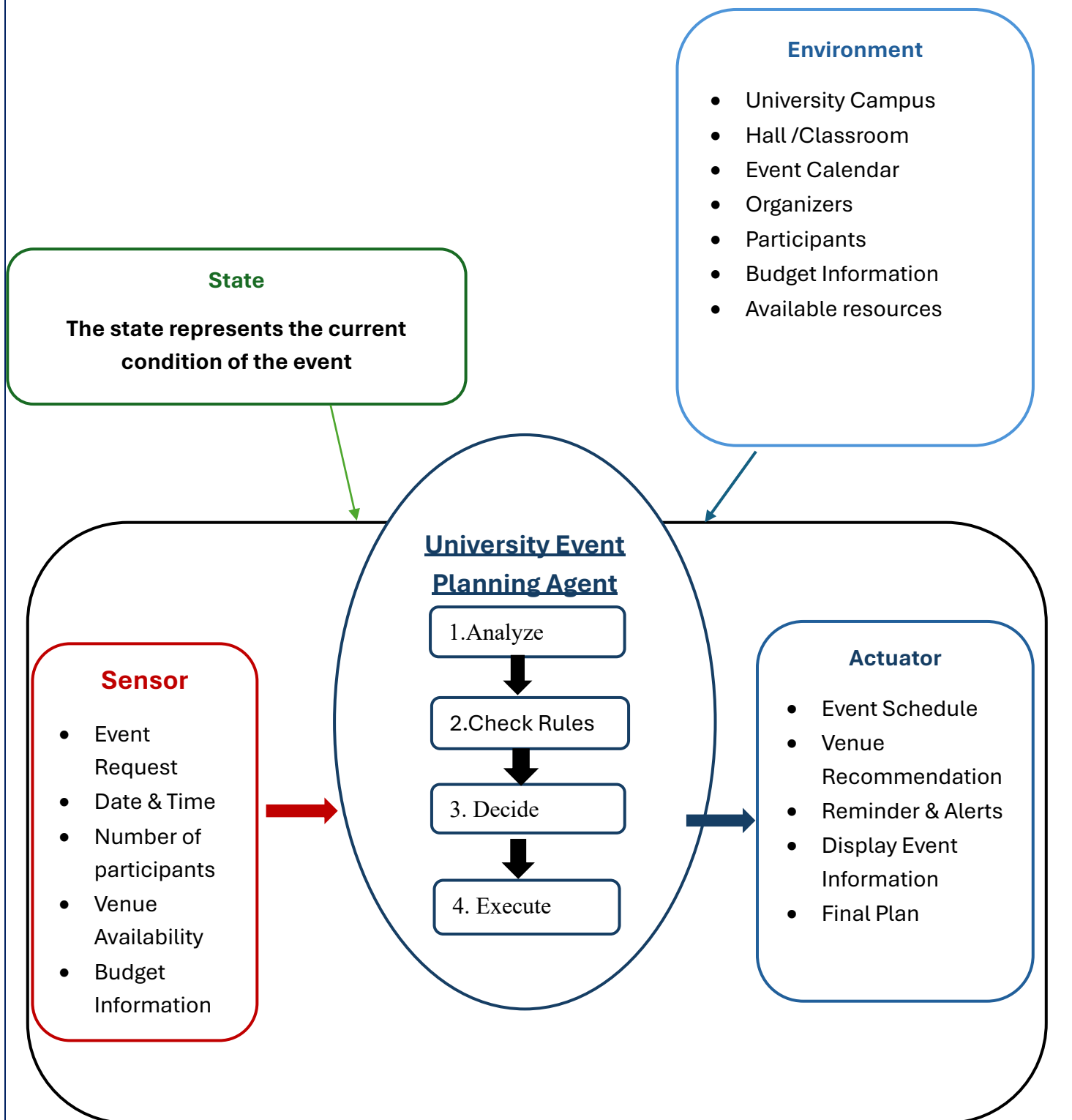
Environment:

The environment includes the university campus, available halls and classrooms, event schedule, event organizers, participants, budget details, and resources needed for the event.

State:

The state shows the present situation of the event, including the chosen venue, event schedule, expected participants, remaining budget, and available resources.

GRAPHICAL REPRESENTATION



3. Evaluate in the Context of PEAS

Performance Measure:

The agent should create an effective event plan, save time, prevent schedule clashes, use resources efficiently, and keep the event within the given budget.

Environment:

The university setting where events are planned, including halls, schedules, participants, organizers, and available resources.

Actuators:

The agent provides event plans, recommendations, notifications, reminders, and alerts.

Sensors:

The agent receives information about the event, such as event type, date, time, participants, venue, budget, and required resources.

4. Identify Environmental Properties

Partially Observable

The agent may not have complete information about participant preferences, venue availability, resource availability, or unexpected changes. Therefore, the environment is partially observable.

Stochastic

Event planning involves uncertainty. Venue availability, attendance, costs, and other conditions may change, so the outcome cannot always be predicted with certainty.

Sequential

The environment is sequential because one decision can affect later decisions. For example, selecting a date can affect venue availability, guest availability, and the event schedule.

Dynamic

The environment is dynamic because event conditions can change during the planning process, such as budget, venue availability, number of participants, or event requirements.

Mostly Discrete

The environment is mostly discrete because the agent makes choices from specific options, such as event type, venue, date, time, and available resources.

Single-Agent

The environment is **single-agent** because one AI agent performs the event-planning and decision-making tasks. Students, faculty, administrators, and vendors are stakeholders, not separate AI agents.

Partially Known

The environment is partially known because the agent has information about event requirements, university rules, available resources, and constraints, but it cannot know every future change or unexpected situation in advance.